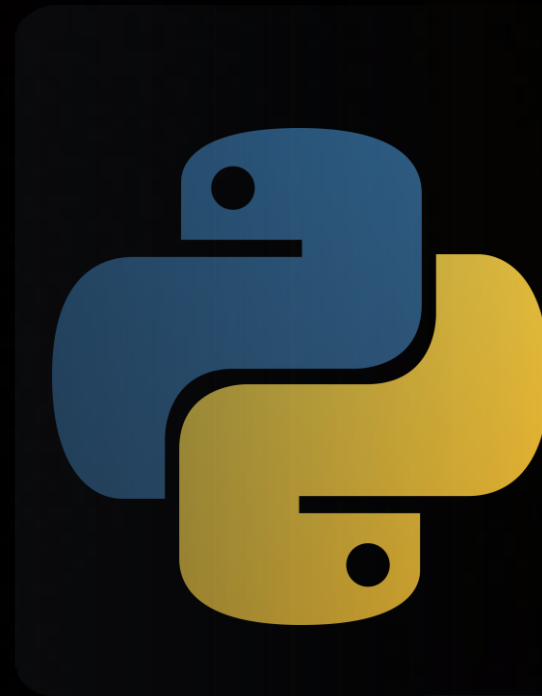


AI based Oil spill detection in marine bodies using SAR

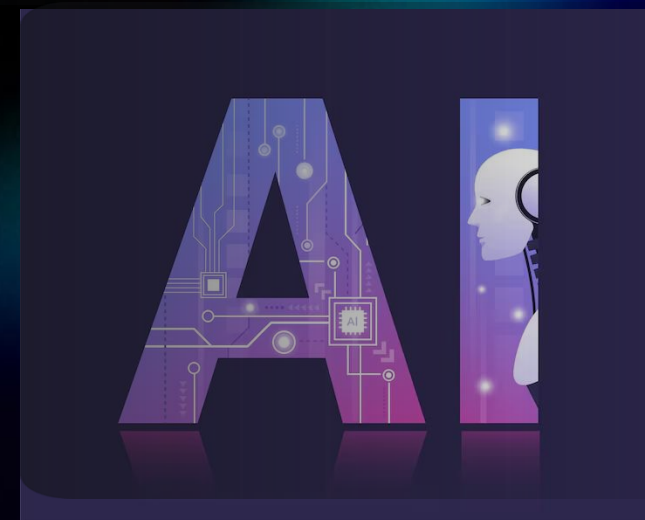
Presented for PBL ICML review 3

OceanSight AI
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Team members

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Problem Statement

01

Oil spills in oceans pose serious threats to marine ecosystems, coastal regions, fisheries, and maritime activities. Detecting spills early and accurately is challenging due to their large spatial extent, varying appearance in satellite imagery, and interference from clouds, waves, and other surface features.

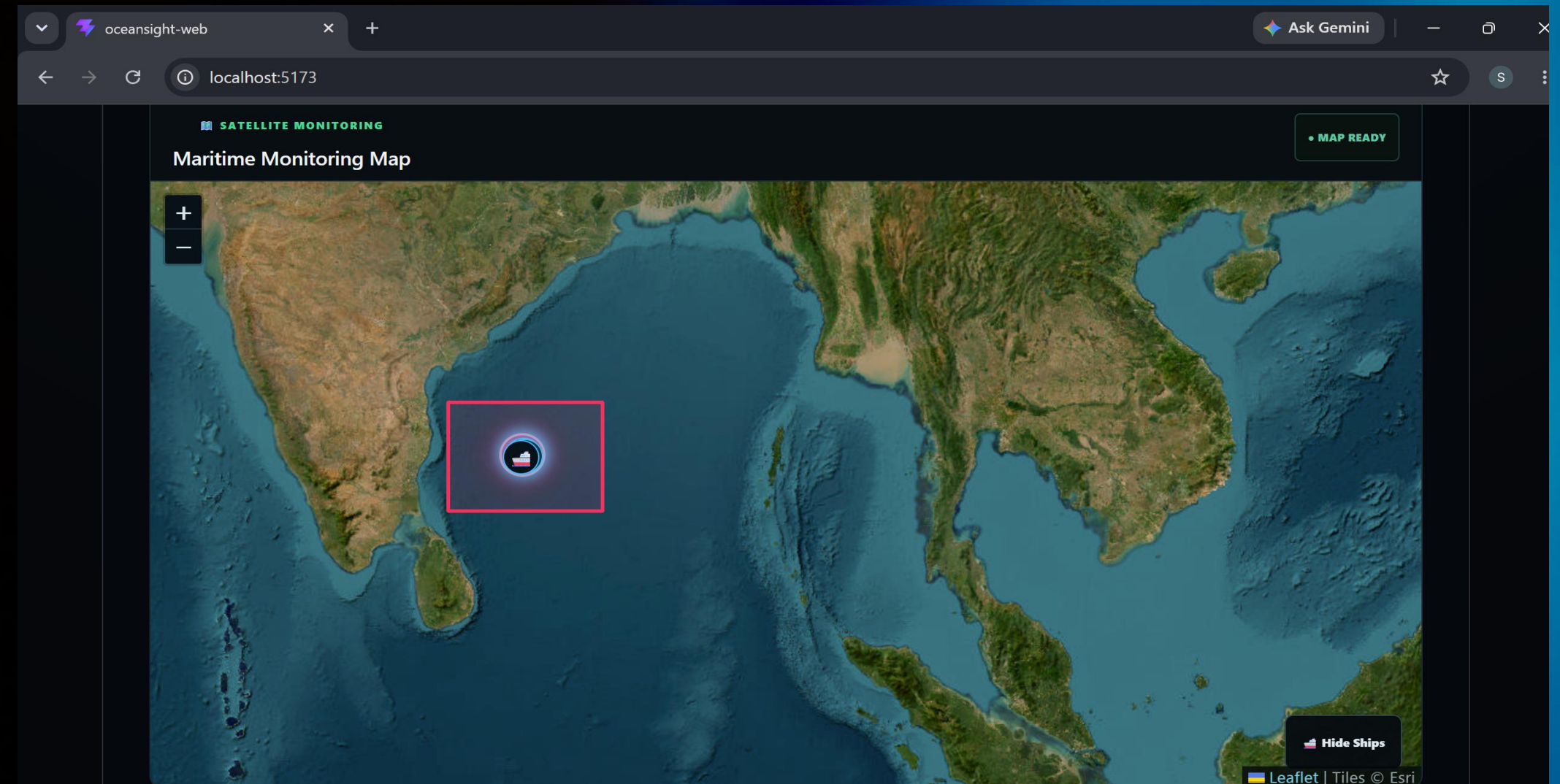


02

Existing monitoring methods often rely on manual analysis or provide limited information beyond detection. There is a need for an **AI-powered system** that can automatically detect oil spills from satellite images, identify their extent and location, and provide actionable information for rapid response and environmental protection.

Our Solution

From satellite imagery to actionable intelligence — detect, map, and monitor oil spills in one platform.



AI-Powered Spill Detection

01

Our deep-learning model analyzes satellite imagery to automatically detect and identify potential oil spills with high accuracy.

Spill Mapping & Analysis

02

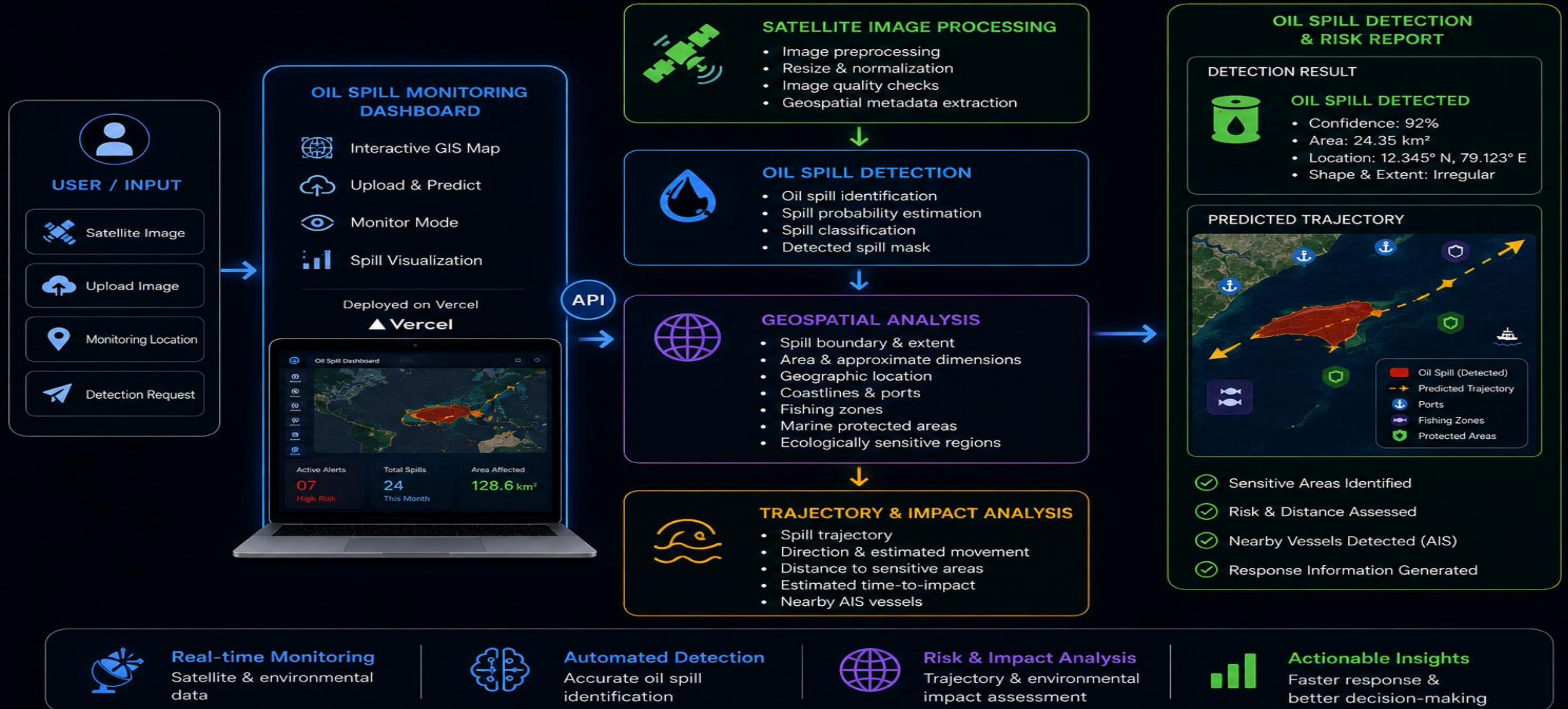
The system identifies the spill boundary and estimates its **location, area, dimensions, and shape**, providing a clear view of the affected region.

Intelligent Environmental Monitoring

03

We combine spill information with coastlines, **ports, fishing zones, protected areas, and predicted movement** to assess environmental risk and support faster response.

System Architecture



Tech Stack

01

AI / ML

Python, TensorFlow, Keras, and MobileNetV2 are used to process satellite imagery, detect oil-spill patterns, and classify potential spill regions with confidence scores

03

GIS & Mapping

Leaflet.js, OpenStreetMap, and GeoJSON enable interactive maps, spill-boundary visualization, geographic positioning, area estimation, and predicted trajectory mapping.

02

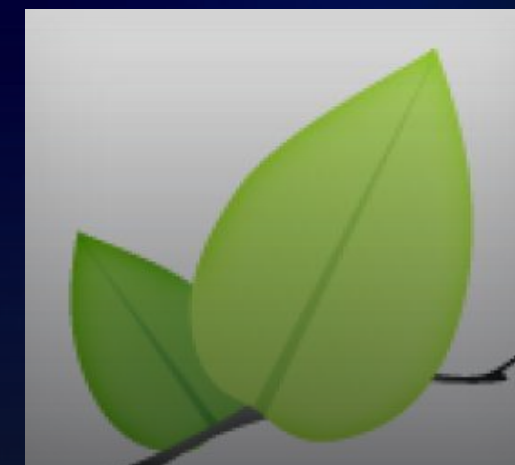
Backend & FAST API

FastAPI, Uvicorn, and Python handle model inference, image processing, API communication, and transfer of detection results between the frontend and backend.

04

Database & Deployment

MongoDB stores spill records, geospatial information, and monitoring data, while Vercel and Render provide scalable deployment for the frontend and backend.



Results Achieved

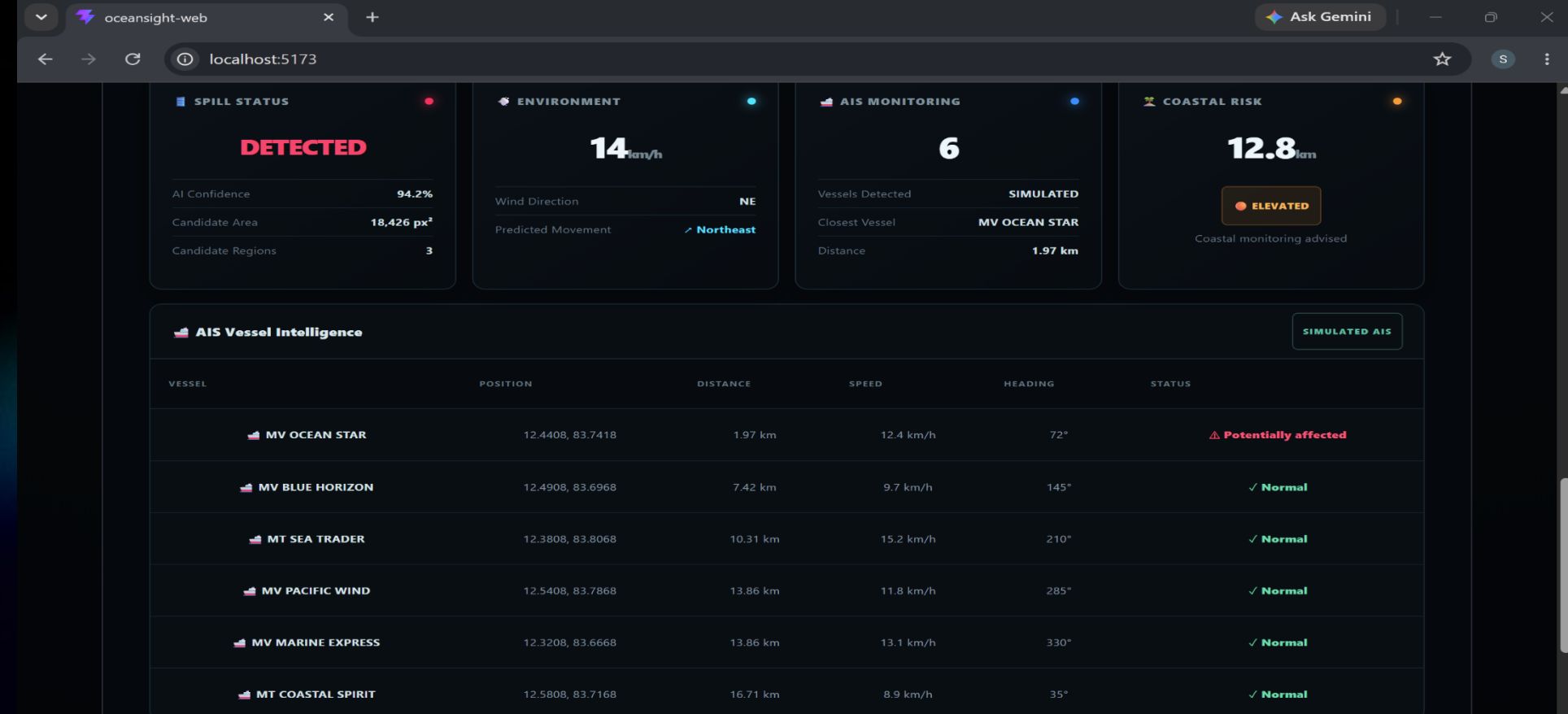
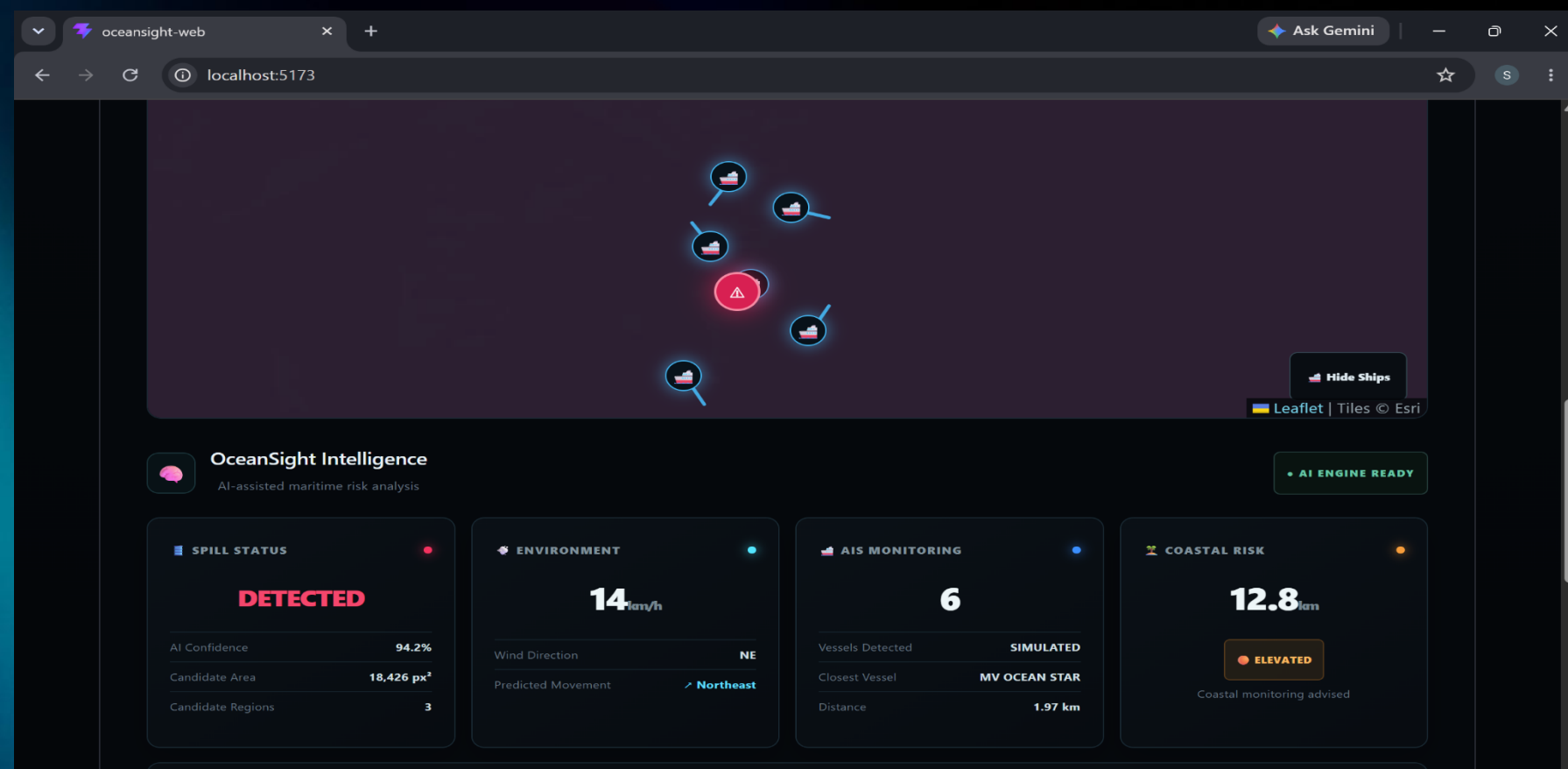
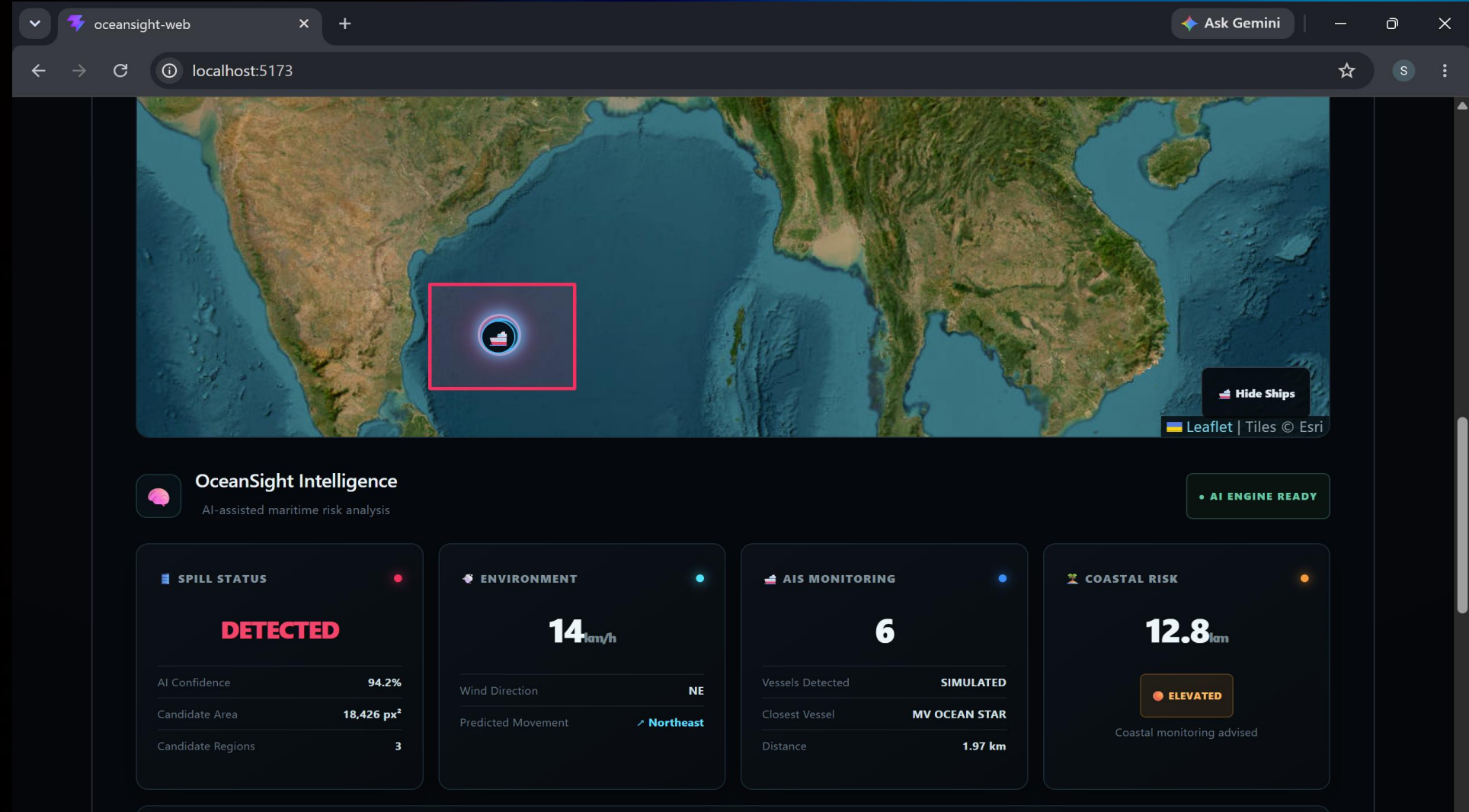
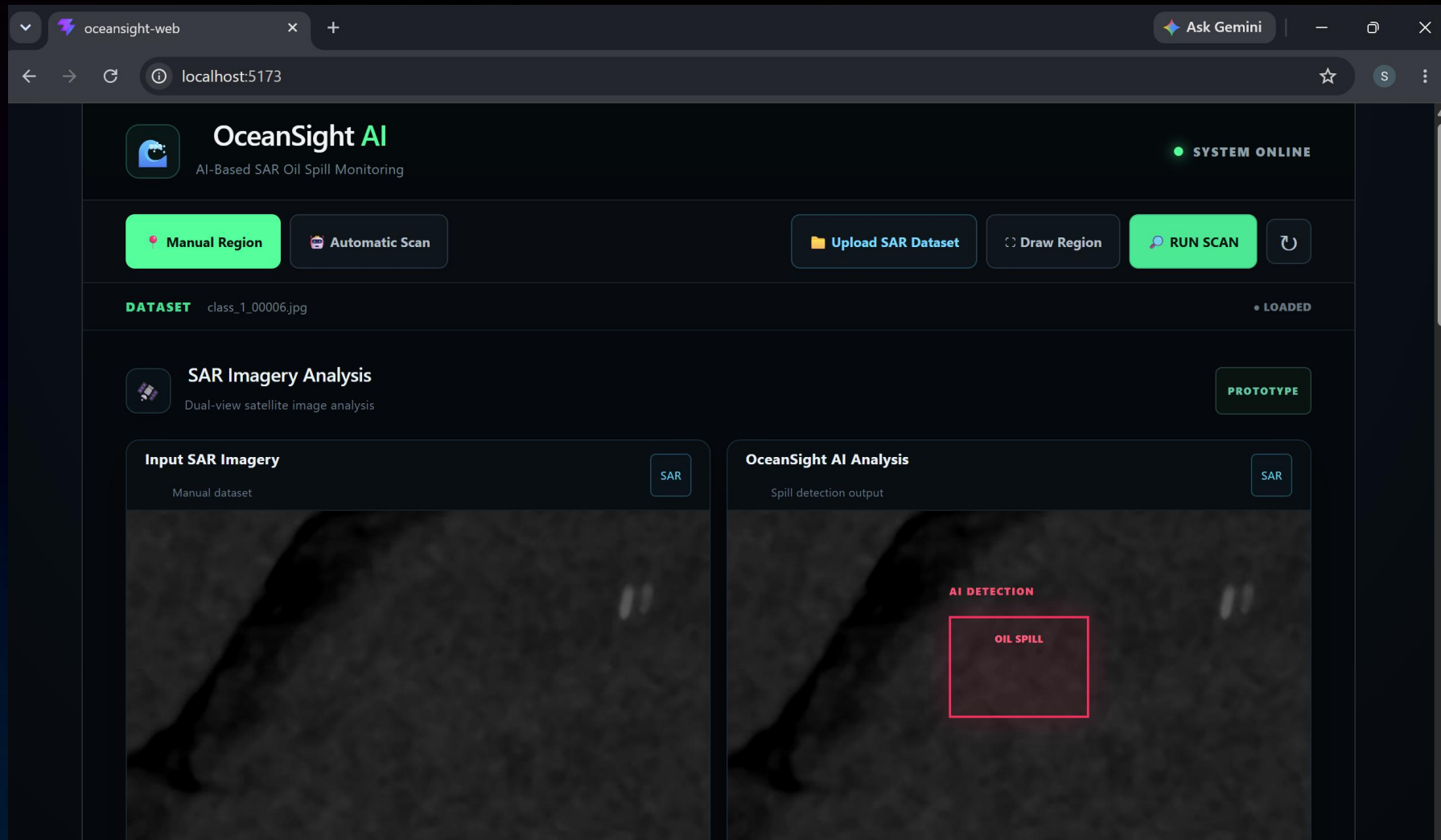
Github repository

<https://github.com/Simonjaques/Oil-Spill-Detection>

Project Demo

<https://drive.google.com/file/d/1lppXnO3fAiJauIDDclq8PTSvtYjWOi09/view?usp=sharing>

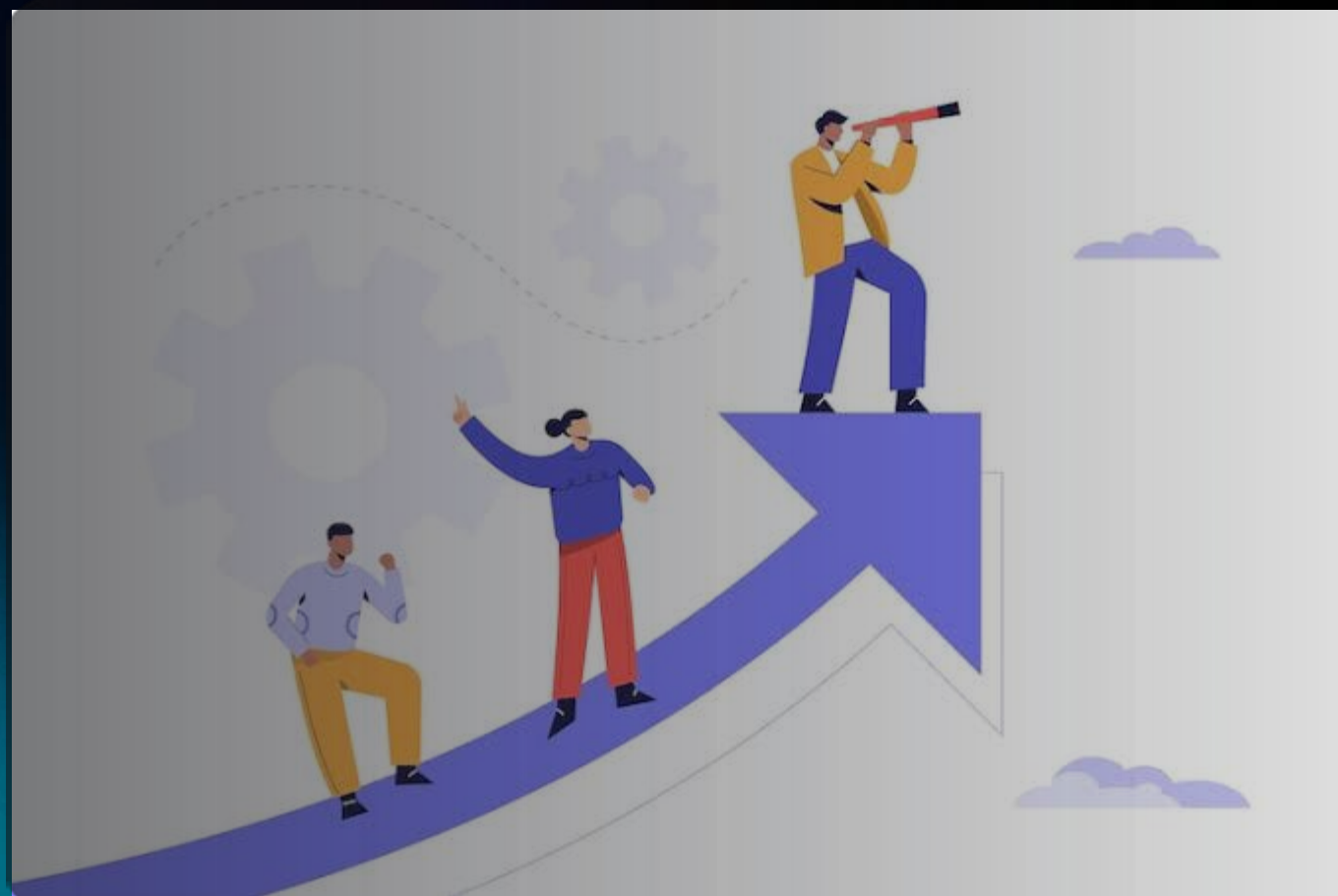




Literature survey

S.No	Authors & Year	Research / Method	Key Findings	Research Gap
1	K. Zeng & Y. Wang (2020)	Deep CNN for oil-spill detection from spaceborne SAR images	Demonstrated deep CNN-based automatic detection and showed the potential of semantic segmentation for SAR oil-spill identification. (MDPI)	Focuses mainly on image-level detection/segmentation , with limited integration of broader geographic risk information.
2	F. M. Bianchi, M. M. Espeseth & N. Borch (2020)	Large-scale deep-learning detection and categorization of oil spills from SAR images	Developed a segmentation-based deep-learning framework capable of large-scale oil-spill detection and categorization. (MDPI)	Primarily addresses detection and categorization , rather than providing an end-to-end environmental monitoring and response platform.
3	M. Shaban et al. (2021)	Deep-learning framework for oil-spill detection from SAR data	Used deep learning to distinguish oil spills from sea-surface features and look-alikes in SAR imagery. (MDPI)	Detection performance can still be affected by look-alike phenomena and varying marine conditions ; operational GIS-based impact assessment is not the main focus.
4	X. Huang, B. Zhang, W. Perrie, Y. Lu & C. Wang (2022)	Faster R-CNN for marine oil-spill detection using C-band SAR	Achieved 89.23% precision and 89.14% recall and demonstrated fast end-to-end detection validated with optical satellite observations. (ScienceDirect)	Concentrates on detecting spill objects , without extending the output into detailed environmental risk, sensitive-area proximity, or trajectory analysis.
5	R. Hasimoto-Beltran et al. (2023)	Multi-channel deep semantic segmentation using SAR, variance and wind-field data	Combined multiple input channels to improve segmentation of complex oil-spill shapes and look-alikes. (ScienceDirect)	Requires additional environmental inputs and focuses on segmentation , leaving scope for a more accessible platform combining detection with geographic decision support.

Future Scope



Real-Time Satellite Monitoring

Integrating continuous satellite data feeds to automatically detect newly emerging oil spills and track their development over time.

Intelligent Spill Trajectory Prediction

Incorporating real-time wind, ocean-current, and wave data to dynamically predict spill movement and identify areas likely to be affected.

Automated Environmental Alerts

Generating instant alerts when detected spills approach coastlines, fishing zones, ports, or marine protected and ecologically sensitive areas.

Global Scalability

Expanding the GIS and backend framework to support large-scale ocean monitoring across different regions and enable long-term historical spill analysis.

Business Model



Government & Environmental Agencies

Providing oil-spill monitoring and risk-assessment solutions to coastal authorities for faster detection, response, and environmental protection.



Maritime & Port Operators

Offering monitoring services to ports, shipping companies, and offshore industries to identify nearby spills and minimize operational and environmental risks.



Subscription-Based Platform

Providing tiered plans with real-time alerts, GIS analysis, trajectory prediction, historical reports, and advanced monitoring features.

PBL Presentation

Thank You

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